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Narula Institute of Technology
An Autonomous Institute under MAKAUT
2024
END SEMESTER EXAMINATION - ODD 2024
IT101 - PROGRAMMING FOR PROBLEM SOLVING

TIME ALLOTTED: 3Hours**FULL MARKS: 70**

Instructions to the candidate:

Figures to the right indicate full marks.

Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Group A
(Multiple Choice Type Questions)

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

- 1.i) Static memory allocation allocates memory during (1) CO3 BL3
a) compile time
b) run time
c) load time
d) link time
- 1.ii) If a variable is declared before the main then (1) CO3 BL3
a) it is a logical error
b) it is a syntax error
c) it is a local variable
d) it is a global variable
- 1.iii) Resolution of externally defined symbols is performed by (1) CO4 BL3
a) linker
b) loader
c) compiler
d) assembler
- 1.iv) What is the binary equivalent of the decimal number 25? (1) CO1 BL2
a) 11001
b) 10011
c) 10101
d) 11100
- 1.v) What is the correct syntax to read an integer using scanf? (1) CO2 BL3
a) scanf("%d", int);
b) scanf("%d", &int);
c) scanf("%d", &var);
d) scanf("%d", *var);

- 1.vi) If p is an integer pointer with a value 1000 in a 32-bit system, then what will the value of p + 5 be? (1) CO5 BL4
- 1005
 - 1010
 - 1004
 - None of these
- 1.vii) What is the size of the int data type in C (on a 32-bit system)? (1) CO3 BL1
- 2 bytes
 - 4 bytes
 - 8 bytes
 - 16 bytes
- 1.viii) Recursion is a process where a function _____. (1) CO3 BL1
- Calls itself
 - Returns no value
 - Executes in a loop
 - Requires a parameter
- 1.ix) How many times does the following loop execute?
for (int i = 0; i < 5; i++) (1) CO1 BL1
- 4
 - 5
 - 6
 - Infinite
- 1.x) _____ is more space effective (1) CO5 BL1
- structure
 - union
 - structure and union
 - none of the above
- 1.xi) sqrt() function's prototype declaration is written within (1) CO4 BL1
- stdio.h
 - stdlib.h
 - math.h
 - conio.h
- 1.xii) breaks are used with (1) CO2 BL1
- loop statements
 - if-else statements
 - switch case statements
 - all of the above

Group B

(Short Answer Type Questions)

(Answer any three of the following) 3x5=15

2. List the main components of modern digital computer system. (5) CO1 BL1

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| 3. Compare break and continue statements in a C-program with example | (5) | CO3 | BL4 |
| 4. Implement array using dynamic memory allocation. | (5) | CO4 | BL3 |
| 5. Differentiate call by value and call by reference with an example. | (5) | CO3 | BL4 |
| 6. Explain the differences between algorithm and flowchart. | (5) | CO1 | BL2 |

Group C
(Long Answer Type Questions)
(Answer any three of the following) 3x15=45

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| 7. Answer all the questions | (15) | | |
| a) Explain the different types of loops in C and provide examples for each (for, while, do-while). | (6) | CO3 | BL2 |
| b) Write a C program that calculates the sum of all even numbers up to a given integer N. | (5) | CO3 | BL6 |
| c) Present an algorithm to find the smallest among three positive integers. | (4) | CO1 | BL3 |
| 8. Answer all the questions | (15) | | |
| a) Implement a C program that uses structures to store and print information about a set of students (name, age, and grade). | (8) | CO4 | BL3 |
| b) Analyze how to use pointers to access the structure members. | (7) | CO4 | BL4 |
| 9. Answer all the questions | (15) | | |
| a) Write a program in C to calculate the length of a string using a pointer. | (5) | CO4 | BL6 |
| b) Write a program in c to find all the Armstrong number between a range(user input). | (4) | CO3 | BL6 |
| c) State the output of the given codes. | (6) | CO2 | BL5 |

a) <pre>int main () { int a, b; a = b = 3; b = a++; printf("%d %d %d %d", a++, --b, ++a, b--); }</pre>	b) <pre>int main() { int m = -10, n = 20; n = (m < 0) ? 0 : 1; printf("%d %d", m, n); printf("\n%d", sizeof(m)); }</pre>	c) <pre>int main() { int marks[] = { 70, 55, 89, 80 }; int *p = &marks[0]; *p++; printf("%d ", *p); *p = *p+2; printf("%d ", *p); }</pre>
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| 10. Answer the following questions. | (15) | | |
| a) Explain with example how to read from and write to a file. | (5) | CO5 | BL2 |
| b) Design a program to find the no of occurrences of a given word in a given file. | (5) | CO5 | BL6 |

- c) Demonstrate the differences between r+ and w+ mode of the file operation. (5) CO5 BL3
11. Write short note (any three): (15) CO5 BL5
- a) Header file
 - b) Type casting
 - c) Binary file and Text file
 - d) Universal logic gates
 - e) Macro
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